University of British Columbia
Department of Computer Science

CPSC 304 Project Cover Page

Milestone #: 3

Date: Oct 23, 2024

Group Number: 43

Name	Student Number	CS Alias (Userid)	Preferred E-mail Address
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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)
In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

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Project Summary

Our project is a communication application modeling user interactions through messages in community spaces called servers. The application also includes premium plans, calendars, events, profile avatars, and assigning roles to accounts such as server administrator and general members.

URL to Group 43 Repository:

https://github.students.cs.ubc.ca/CPSC304-2024W-T1/project_c6n7z_p3z3e_w2z8i

Timeline and Task Breakdown:

Part 1 - Backend

1.1 Create tables: [Oct 26, 2024]

1. Payment
2. Tier
3. PremiumPlan
4. Location
5. Avatar
6. UserAccount
7. Calendar
8. Event
9. Server
10. Channel
11. GeneralMember
12. Administrator
13. Message
14. PostedTo
15. Joins

Created in M2, edited to work in Oracle by: Oct 26 2024

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1.2 Queries: [Oct.28 - Nov.10]

Task Timeline:

- Initial test queries (~2 per requirement to ensure it works) to be implemented by: [Oct. 28]
- All below listed queries to be implemented by: [Nov 2]
- All additional queries to be planned by: [Nov 3]
- All additional queries to be implemented by: [Nov 10]

Task assignments per group member:

- Ananya: Insert, Selection, Aggregation with Group By, Division
- Karen: Update, Projection, Aggregation with Having
- Allison: Delete, Join, Nested aggregation with group by

Details of Backend Tasks:

Insert
Message: Contents
User account: all details, AvatarID
Event: all details except eventID
Calendar: CalendarName
If admin: tag, signature, server name
If general member: signature

Update
User account: all details [error check for unique username]
Event: all details except eventID
Calendar: CalendarName
If admin: tag, signature, server name
If general member: tag

Delete
User account: all details (username, password and region are either all present or all deleted meaning user has deleted account)
Event: all details except eventID (if event exists, eventName must exist)
Calendar: delete calendar (if calendar exists, calendarName must exist)
If admin: tag, signature, server name
If general member: tag

Selection
Server: selection of list of servers based on criteria and filters on attributes such as backgroundColor, Name.

Projection
PremiumPlan: view only specific columns like Tier, Price, theme, etc.
Joins - view all users part of a server (will have to use Join query as well as projection)

Join
PremiumPlan: join on keys for Tier, PaymentInterval, PremiumPlan to allow user to view full plan details.

Aggregation with Group By
Events by month: aggregate count of events grouped by calendar month to view calendar activity Count of users group by region and country

Aggregation with Having
Server metrics: aggregate the channels having count more than a certain number of users

Nested Aggregation with Group By
Server metrics: aggregate to find maximum of average of messages sent in server channel group by date, having above a certain value to filter for high activity channels

Division
ViewActiveServer - view only servers where by a specific user has commented in every single channel
ViewEventsinAllCalendars - view events that belong to all of a user's calendars

Part 2 - Frontend

Task Timeline:

- Create Pages Layout: [Oct 30]
- Full pages without backend connection: [Nov 10]
- Connecting to backend (adding functionality to buttons, checklists, text boxes etc.): [Nov 20]
- Error checking: [Nov 25]
- Final tests and verification: [Nov 27]

Task assignments per group member:

- Ananya: Home, Plan, login, create, profile page
- Karen: Calendar, create event, edit event
- Allison: Server Page, create server, admin metrics, join another server

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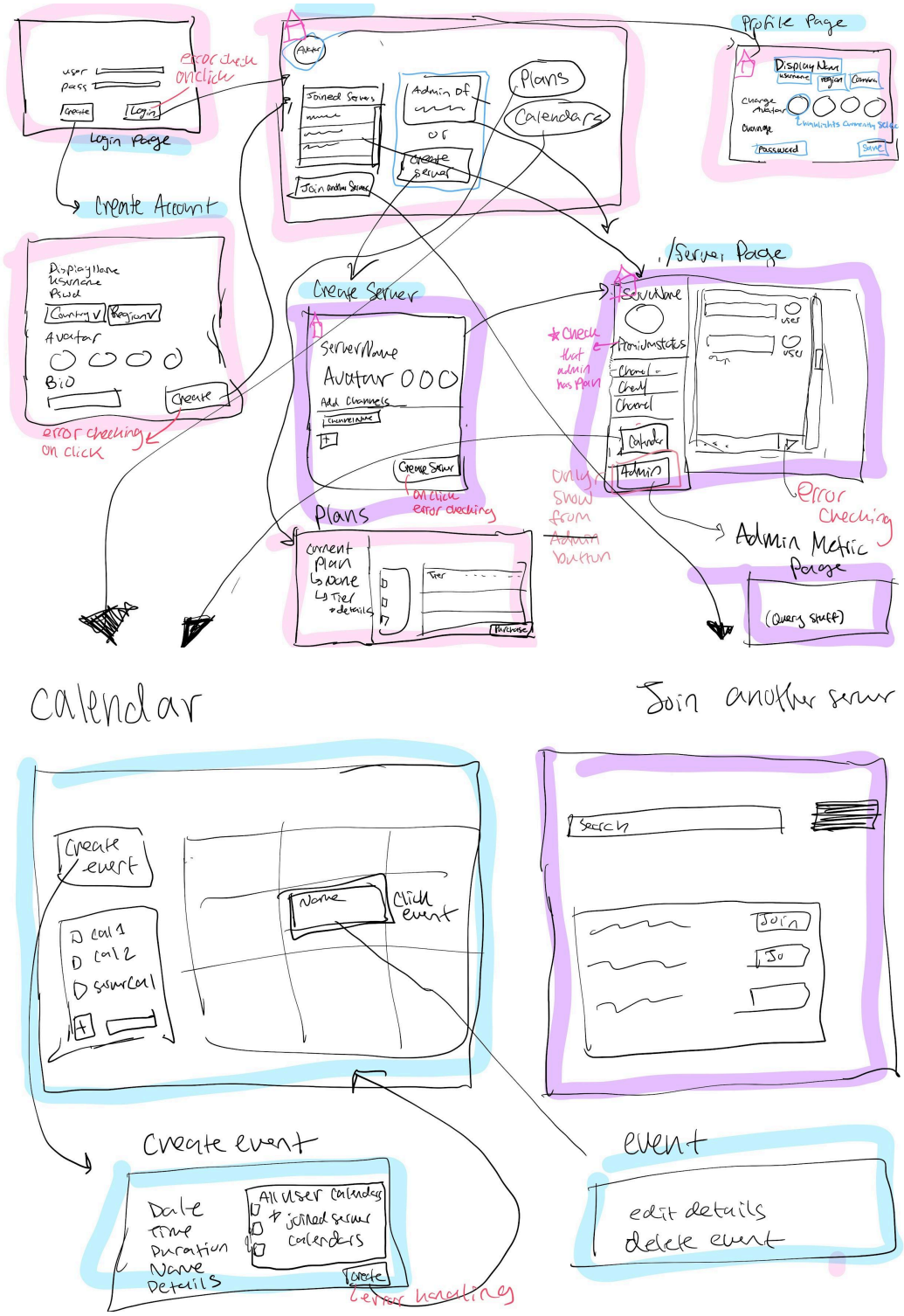
DRAFT OF FRONTEND:



FINAL DEADLINE: IMPLEMENTED PROJECT DUE NOVEMBER 29 10:00 PM